STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE – 12

RANDOM VARIABLES

CONTINUOUS RANDOM VARIABLE

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THE PROBABILITY DENSITY FUNCTIONS

A random variable X is said to be continuous if there is a function $f(x)$, called the probability density function, such that.

① $f(x) \geq 0$ for all x.

② $\int_{-\infty}^{+\infty} f(x)\,dx = 1$

③ $P(a \leq x \leq b) = \int_{a}^{b} f(x)\,dx$.

Notice that for a continuous random variable X

① $P(X = a) = \int_{a}^{a} f(x)\,dx = 0$

② $P(a \leq x \leq b) = P(a < x \leq b) = P(a \leq x < b) = P(a < x < b)$

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EXAMPLES OF PROBABILITY DENSITY FUNCTIONS

EXAMPLE ① Let x be a random variable having the function (p.d.f).

$$f(x) = \begin{cases} cx, & 0 \le x \le 2 \\ 0, & \text{otherwise.} \end{cases}$$

Find

(a) the value of c.

(b) $P(x < 0.5)$

(c) $P(x > 1)$

(d) $P(\frac{1}{2} < x < \frac{3}{2})$

(e) $P(x = 1)$

SOLUTION (a) WKT $\int_{-\infty}^{+\infty} f(x)\,dx = 1$

$$\Rightarrow \int_0^2 cx\,dx = 1$$

$$c\left|\frac{x^2}{2}\right|_0^2 = 1 \;,\; 2c = 1 \;,\; \boxed{c = \frac{1}{2}}$$

(b) $P(x < 0.5) = \int_0^{0.5} \frac{1}{2}x\,dx = \left|\frac{x^2}{4}\right|_0^{0.5} = 0.0625$

(c) $P(x > 1) = \int_1^2 \frac{1}{2}x\,dx = \left|\frac{x^2}{4}\right|_1^2 = 1 - \frac{1}{4} = \frac{3}{4} = 0.75$

(d) $P(\frac{1}{2} < x < \frac{3}{2})$

$$= \int_{\frac{1}{2}}^{3/2} \frac{1}{2}x\,dx = \left|\frac{x^2}{4}\right|_{1/2}^{3/2} = \frac{1}{2} = 0.5$$

(e) $P(x = 1) = 0$

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EXAMPLE-2 X is the delay, in hours of a flight from chicago, where

$$f(x) = 0.2 - 0.02x \quad , \quad 0 \le x \le 10$$

Find

(a) the probability that the delay will be less than four hours.

(b) the probability that the delay will be between two and six Hours.

SOLUTION Here $f(x) = 0.2 - 0.02x \quad , \quad 0 \le x \le 10$

(a) $P(0 \le x \le 4) = \int_0^4 (0.2 - 0.02x)\,dx$

$$= \left| 0.2x - 0.02\frac{x^2}{2} \right|_0^4$$

$$= ((0.2)(4) - (0.01)(4)^2) - 0$$

$$= 0.64$$

(b) $P(2 \le x \le 6) = \int_2^6 (0.2 - 0.02x)\,dx$

$$= \left| 0.2x - 0.01x^2 \right|_2^6$$

$$= [(0.2)(6) - (0.01)(36)] - [(0.2)(2) - (0.01)(4)]$$

$$= (0.84) - (0.36) = 0.48$$

EXAMPLE③ A continuous random variable has p.d.f

$$f(x) = \begin{cases} kx, & 0 \le x < 2 \\ k(4-x), & 2 \le x < 4 \\ 0, & \text{otherwise.} \end{cases}$$

Find

(a) $P(x < 1)$ (b) $P(x > 3)$ (c) $P(0.5 \le x \le 2.5)$

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First of all we shall find 'k'.

Wkt $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$\int_{0}^{4} f(x)\,dx = 1$

$\int_0^2 kx\,dx + \int_2^4 k(4-x)\,dx = 1$

$k\left|\frac{x^2}{2}\right|_0^2 + k\left|4x - \frac{x^2}{2}\right|_2^4 = 1$

$\frac{k}{2}(4) + k(8-6) = 1$

$2k + 2k = 1 \Rightarrow 4k = 1 \Rightarrow \boxed{k = \frac{1}{4}}$

Now

(a) $P(x<1) = \int_0^1 \frac{1}{4}x\,dx$

$= \frac{1}{4}\left|\frac{x^2}{2}\right|_0^1 = \frac{1}{8}(1-0) = \frac{1}{8}$

(b) $P(x>3) = \int_3^4 \frac{1}{4}(4-x)\,dx$

$= \frac{1}{4}\left|4x - \frac{x^2}{2}\right|_3^4 = \frac{1}{4}\left(8 - 12 + \frac{9}{2}\right)$

$= \frac{1}{4}\left(-4 + \frac{9}{2}\right) = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$

(c) $P(0.5 \le x \le 2.5)$

$= \int_{0.5}^{2.5} f(x)\,dx$

$= \int_{0.5}^{2} \frac{1}{4}x\,dx + \int_{2}^{2.5} \frac{1}{4}(4-x)\,dx$

$= \frac{1}{4}\left|\frac{x^2}{2}\right|_{0.5}^{2} + \frac{1}{4}\left|4x - \frac{x^2}{2}\right|_{2}^{2.5}$

$= \frac{1}{8}(3.75) + \frac{1}{4}(0.875) = 0.6875.$

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EXAMPLE ④ The continuous random variable X has p.d.f, $f(x)$ where

$$f(x) = \begin{cases} k(x+2)^2 & , -2 \le x < 0 \\ 4k & , 0 \le x \le \frac{4}{3} \\ 0 & , \text{otherwise.} \end{cases}$$

(a) Find the value of k.

(b) $P(-1 \le x \le 1)$

(c) $P(x > 1)$

SOLUTION (a) WKT $\int_{-\infty}^{\infty} f(x)\,dx = 1$, $\int_{0}^{\frac{4}{3}} f(x)\,dx = 1$

$$\Rightarrow \int_{-2}^{0} k(x+2)^2\,dx + \int_{0}^{\frac{4}{3}} 4k\,dx = 1$$

$$k\left|\frac{(x+2)^3}{3}\right|_{-2}^{0} + 4k\,|x|_{0}^{4/3} = 1$$

$$k\left[\frac{8}{3} - 0\right] + 4k\left[\frac{4}{3} - 0\right] = 1$$

$$\frac{8k}{3} + \frac{16k}{3} = 1$$

$$\frac{24k}{3} = 1 \Rightarrow \boxed{k = \frac{1}{8}}$$

(b) $P(-1 \le x \le 1) = \int_{-1}^{1} f(x)\,dx$

$$= \int_{-1}^{0} \frac{1}{8}(x+2)^2\,dx + \int_{0}^{1} 4 \cdot \frac{1}{8}\,dx$$

$$= \frac{1}{8}\cdot\left|\frac{(x+2)^3}{3}\right|_{-1}^{0} + \frac{1}{2}|x|_{0}^{1}$$

$$= \frac{1}{24}(8-1) + \frac{1}{2}(1-0) = \frac{7}{24} + \frac{1}{2} = \frac{19}{24}$$

(c) $P(x > 1) = \int_{1}^{\frac{4}{3}} 4k\,dx$

$$= \frac{4}{8}\int_{1}^{\frac{4}{3}} 1\,dx = \frac{1}{2}|x|_{1}^{4/3} = \frac{1}{2}\left(\frac{4}{3} - 1\right) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

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EXAMPLE (5) A continuous random variable x has the Probability density function.

$$f(x) = \begin{cases} x/2 , & 0 \le x < 1 \\ (3-x)/4 , & 1 \le x < 2 \\ 1/4 , & 2 \le x < 3 \\ (4-x)/4 , & 3 \le x < 4 \\ 0 , & \text{otherwise.} \end{cases}$$

Compute.

(a) $P(x < 2)$

(b) $P(x \ge 3)$

(c) $P(1 < x < 3)$

(d) $P(|x| < 1.5)$

(e) $P(|x| > 1.25)$

SOLUTION

(a) $P(x<2) = \int_0^2 f(x)\,dx = \int_0^1 \frac{x}{2}\,dx + \int_1^2 \frac{3-x}{4}\,dx$

$= \frac{1}{2}\left|\frac{x^2}{2}\right|_0^1 + \frac{1}{4}\left|3x - \frac{x^2}{2}\right|_1^2$

$= \frac{1}{4}(1-0) + \frac{1}{4}\left[(6-2) - (3-\frac{1}{2})\right]$

$= \frac{1}{4} + \frac{1}{4}\left[4 - \frac{5}{2}\right]$

$= \frac{1}{4} + \frac{1}{4} \times \frac{3}{2} = \frac{1}{4} + \frac{3}{8} = \frac{5}{8}$

(b) $P(x \ge 3) = \int_3^4 \frac{4-x}{4}\,dx$

$= \frac{1}{4} \cdot \left|4x - \frac{x^2}{2}\right|_3^4$

$= \frac{1}{4}\left[(16-8) - (12 - \frac{9}{2})\right]$

$= \frac{1}{4}\left[8 - \frac{15}{2}\right] = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$

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(c) $P(1<X<3)$

$$= \int_1^3 f(x)\,dx = \int_1^2 \frac{3x}{4}\,dx + \int_2^3 \frac{1}{4}\,dx.$$

$$= \frac{1}{4}\left|3x - \frac{x^2}{2}\right|_1^2 + \frac{1}{4}\left|x\right|_2^3$$

$$= \frac{1}{4}\left[(6-2)-(3-\tfrac{1}{2})\right] + \frac{1}{4}\left[3-2\right]$$

$$= \frac{1}{4}\left[4-\frac{5}{2}\right] + \frac{1}{4}$$

$$= \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$$

(d) $P(|X|<1.5) = P(-1.5<X<1.5)$

$$= \int_{-1.5}^{1.5} f(x)\,dx = \int_{-1.5}^{0} 0\,dx + \int_0^1 \frac{x}{2}\,dx + \int_1^{1.5} \frac{3-x}{4}\,dx$$

$$= \frac{1}{4}\left|x^2\right|_0^1 + \frac{1}{4}\left|3x - \frac{x^2}{2}\right|_1^{1.5}$$

$$= \frac{1}{4}(1) + \frac{1}{4}(3.375 - 2.5)$$

$$= \frac{1}{4} + \frac{1}{4}(0.875) = 0.46875$$

(e) $P(|X|>1.25) = P(X<-1.25) + P(X>1.25)$ ~~$= 0$~~

$$= \int_{1.25}^{4} f(x)\,dx$$

$$= \int_{1.25}^{2} \left(\frac{3-x}{4}\right)dx + \int_2^3 \frac{1}{4}\,dx + \int_3^4 \left(\frac{4-x}{4}\right)dx$$

$$= \frac{1}{4}\left|3x - \frac{x^2}{2}\right|_{1.25}^{2} + \frac{1}{4}\left|x\right|_2^3 + \frac{1}{4}\left|4x - \frac{x^2}{2}\right|_3^4$$

$$= \frac{1}{4}(4 - 2.96875) + \frac{1}{4}(1) + \frac{1}{4}(8-7.5)$$

$$= \frac{1}{4}\left[2.53125\right] = 0.6328125.$$

or $P(|X|>1.25) = 1 - P(|X|<1.25) = 0.6328125.$

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DISTRIBUTION FUNCTION FOR CONTINUOUS RANDOM VARIABLE

The dist. function for a random variable 'x' is defined as $F(b) = P(x \leq b)$

If 'x' is continuous RV with p.d.f $f(x)$, then.

$$F(b) = \int_{-\infty}^{b} f(x)\,dx$$

IMPORTANT RESULTS

① $f(x) = F'(x)$ ie $\text{P.d.f} = \frac{d}{dx}(\text{CDF})$

② $P(a \leq x \leq b) = F(b) - F(a)$

③ Median.

If 'm' is the median, then for $f(x)$ defined for $a \leq x \leq b$. $F(m) = 0.5$, solve and find 'm'.

④ Lower quartile

If 'q_1' is the lower quartile. then to find 'q_1' put

$$F(q_1) = 0.25$$

⑤ Upper quartile

If 'q_3' is the upper quartile, then to find 'q_3', put

$$F(q_3) = 0.75$$

⑥ In general

$$F(n\text{th Percentile}) = \frac{n}{100}$$

⑦ Interquartile Range $= q_3 - q_1$

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EXAMPLES

EXAMPLE ① 'X' is a continuous RV with p.d.f

$$f(x) = \begin{cases} \frac{1}{8}x, & 0 \le x \le 4 \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find CDF

(b) $P(0.3 \le X \le 1.8)$

(c) Median 'm'.

(d) Interquartile Range

SOLUTION

(a) $$F(x) = \int_{-\infty}^{x} \frac{1}{8} x\,dx = \int_{0}^{x} \frac{1}{8} x$$

$$= \left| \frac{x^2}{16} \right|_0^x = \frac{x^2}{16}$$

$$\Rightarrow F(x) = \begin{cases} 0, & x \le 0 \\ x^2/16, & 0 \le x \le 4 \\ 1, & x \ge 4 \end{cases}$$

(b) $P(0.3 \le x \le 1.8) = F(1.8) - F(0.3)$

$$= \frac{(1.8)^2}{16} - \frac{(0.3)^2}{16} = 0.005625$$

(c) Median $F(m) = 0.5$

$$\frac{m^2}{16} = 0.5$$

$$m^2 = 8 \Rightarrow \boxed{m = 2.83}$$

→ Ignore -ve as $0 \le x \le 4$.

(d) Interquartile Range

$$IQR = q_3 - q_1$$

$F(q_1) = 0.25$, $\frac{q_1^2}{16} = 0.25 \Rightarrow q_1 = 2$

$F(q_3) = 0.75$, $\frac{q_3^2}{16} = 0.75 \Rightarrow q_3 = 3.464$

$\Rightarrow IQR = 3.464 - 2 = 1.5$

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EXAMPLE② A continuous random variable 'X' has the density function.

$$f(x) = \begin{cases} \theta e^{-\theta x}, & 0 \le x < \infty \\ 0, & \text{otherwise.} \end{cases}$$

Find CDF.

SOLUTION $F(x) = \int_0^x \theta e^{-\theta x} dx.$

$$= \theta \cdot \left| \frac{e^{-\theta x}}{-\theta} \right|_0^x$$

$$= - \left| e^{-\theta x} \right|_0^x$$

$$= - [e^{-\theta x} - 1]$$

$$F(x) = 1 - e^{-\theta x}$$

do $$F(x) = \begin{cases} 1 - e^{-\theta x}, & x > 0 \\ 0, & x \le 0 \end{cases}$$

EXAMPLE③ 'X' is a continuous random variable with p.d.f $f(x)$ where

$$f(x) = \begin{cases} \frac{x}{3}, & 0 \le x \le 2 \\ -\frac{2x}{3} + 2, & 2 \le x \le 3 \\ 0, & \text{otherwise.} \end{cases}$$

Find the following

(a) CDF

(b) $P(1 \le X \le 2.5)$

(c) Median.

SOLUTION (a) CDF

For CDF we shall integrate two times Here.

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Now $F_1(x) = \int_0^x \frac{x}{3}\,dx = \left|\frac{x^2}{6}\right|_0^x = \frac{x^2}{6}$, $0 \le x \le 2$

Now for $2 \le x \le 3$

$$F_2(x) = F_1(2) + \int_2^x \left(-\frac{2x}{3} + 2\right) dx$$

$$= \frac{4}{6} + \left| -\frac{x^2}{3} + 2x \right|_2^x$$

$$= \frac{2}{3} + \left\{ -\frac{x^2}{3} + 2x + \frac{4}{3} - 4 \right\}$$

$$= \frac{2}{3} - \frac{x^2}{3} + 2x - \frac{8}{3}$$

$$F_2(x) = -\frac{x^2}{3} + 2x - 2$$

So $$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x^2}{6}, & 0 \le x \le 2 \\ -\frac{x^2}{3} + 2x - 2, & 2 \le x \le 3 \\ 1, & x \ge 3 \end{cases}$$

(b) $P(1 \le x \le 2.5) = F(2.5) - F(1)$

$$= \left\{ -\frac{(2.5)^2}{3} + 2(2.5) - 2 \right\} - \left\{ \frac{1}{6} \right\} = \frac{11}{12} - \frac{1}{6} = 0.75$$

(c) For Median $F(m) = 0.5$

$$\frac{m^2}{6} = 0.5 \quad (\text{As Median must be} < 2)$$

$$m^2 = 3$$

$$m = \pm 1.73$$

Reject -ve value as $0 \le x \le 3$

So $\boxed{m = 1.73}$

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EXAMPLE (4) A continuous RV 'X' has the distribution function as

$$F(x) = \begin{cases} 0 & , \ x \le 0 \\ x^3/27 & , \ 0 < x < 3 \\ 1 & , \ x \ge 3 \end{cases}$$

Find

(a) P.d.f of X

(b) $P(X>2)$ using p.d.f and c.d.f.

SOLUTION

(a) $f(x) = \frac{d}{dx} F(x)$

$$\Rightarrow f(x) = \begin{cases} x^2/9 & , \ 0 \le x \le 3 \\ 0 & , \ \text{otherwise.} \end{cases}$$

(b) $P(X>2)$

using P.d.f

$$P(X>2) = \int_2^3 \frac{x^2}{9} dx = \frac{1}{27} \left| x^3 \right|_2^3$$

$$= \frac{1}{27} (27-8) = \frac{19}{27}$$

using c.d.f

$$P(X>2) = P(2<X<3)$$

$$= F(3) - F(2)$$

$$= \frac{(3)^3}{27} - \frac{(2)^3}{27}$$

$$= \frac{27}{27} - \frac{8}{27}$$

$$= \frac{19}{27}$$

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EXAMPLE ⑤ The distribution function for a Random variable 'x' is given by

$$F(x) = \begin{cases} 0 & , x < 0 \\ x/4 & , 0 \le x < 1 \\ x/2 & , 1 \le x < 2 \\ (6x - x^2)/4 & , 2 \le x < 3 \\ 1 & , x \ge 3 \end{cases}$$

Find

(a) P.d.f of x.

(b) $P(1 \le x \le 2)$ using p.d.f and c.D.F.

SOLUTION

(a) $f(x) = \frac{d}{dx} F(x)$

So $f(x) = \begin{cases} 1/4 & , 0 \le x < 1 \\ 1/2 & , 1 \le x < 2 \\ (3-x)/2 & , 2 \le x < 3 \\ 0 & , \text{otherwise} \end{cases}$

(b) $P(1 \le x \le 2)$

using p.d.f

$$P(1 \le x \le 2) = \int_1^2 \frac{1}{2} dx$$

$$= \frac{1}{2} |x|_1^2 = \frac{1}{2}(2-1) = \frac{1}{2}$$

using CDF

$$P(1 \le x \le 2) = F(2) - F(1)$$

$$= \frac{2}{2} - \frac{1}{2}$$

$$= 1 - \frac{1}{2} = \frac{1}{2}$$

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EXPECTATION CRV'S

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EXPECTATION OF CONTINUOUS RANDOM VARIABLES

If 'x' is a continuous random variable, than the Mean or expected value of 'x' is given by

$$E(x) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx.$$

If 'x' is a continuous random variable with density function $f(x)$, than expectation of the function $g(x)$ is given by $E(g(x)) = \int_{-\infty}^{\infty} g(x) \cdot f(x)\, dx.$

Note that

① $E(x) = \int_{-\infty}^{\infty} x f(x)\, dx$

② $E(x^2) = \int_{-\infty}^{\infty} x^2 \cdot f(x)\, dx$

PROPERTIES OF EXPECTATION

If 'x' is a CRV and a, b are constants. than

(a) $E(a) = a$

(b) $E(ax) = a \cdot E(x)$

(c) $E(ax+b) = a \cdot E(x) + b$

VARIANCE AND STANDARD DEVIATION.

If X is a CRV than

1) $Var(x) = E(x^2) - (E(x))^2$

2) $S.D(x) = \sqrt{Var(x)}$

PROPERTIES OF VARIANCE.

If X is a CRV and a, b are constants than

1) $Var(a) = 0$

2) $Var(ax) = a^2 \cdot V(x)$

3) $Var(ax+b) = a^2 \cdot Var(x)$

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EXAMPLES

EXAMPLE-① The continuous random variable x has p.d.f given by

$$f(x) = \begin{cases} k(x+3), & 0 \le x \le 4 \\ 0, & \text{otherwise.} \end{cases}$$

Find the following

(a) the value of k.

(b) $P(x<1)$

(c) $E(x)$

(d) $E(x^2)$

(e) $E(2x+5)$

SOLUTION (a) $c = ?$

WKT $\int_0^4 k(x+3)\,dx = 1$

$$k\left|\frac{x^2}{2}+3x\right|_0^4 = 1$$

$$k\left(\frac{16}{2}+12\right) = 1$$

$$20k = 1$$

$$\boxed{k = \frac{1}{20}}$$

(b) $P(x<1)$

$$= \int_0^1 \frac{1}{20}(x+3)\,dx = \frac{1}{20}\left|\frac{x^2}{2}+3x\right|_1^4$$

$$= \frac{1}{20}\left[(8+12)-\left(\frac{1}{2}+3\right)\right] = \frac{1}{20}\left[20-\frac{7}{2}\right] = \frac{33}{40}$$

(c) $E(x) = \int_0^4 x \cdot \frac{1}{20}(x+3)\,dx = \frac{1}{20}\int_0^4 (x^2+3x)\,dx$

$$= \frac{1}{20}\left|\frac{x^3}{3}+\frac{3}{2}x^2\right|_0^4 = \frac{1}{20}\left(\frac{64}{3}+\frac{48}{2}\right) = \frac{34}{15} = 2.267$$

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(d) $E(x^2)$

$$= \int_0^4 x^2 \cdot \frac{1}{20} \cdot (x+3)\,dx = 6.4$$

(e) $E(2x+5)$

$$= 2E(x)+5 = 2\left(\frac{34}{15}\right)+5 = \frac{143}{15} = 9.53.$$

EXAMPLE-② The continuous random variable x has p.d.f given below

$$f(x) = \begin{cases} x/8, & 0 \le x \le 4 \\ 0, & \text{otherwise.} \end{cases}$$

Find the following

(a) $E(x)$

(b) $E(x^2)$ (c) $Var(x)$ (d) $S.D(x)$ (e) $Var(3x+2)$.

SOLUTION

(a) $$E(x) = \int_0^4 \frac{x}{8} \cdot x \cdot dx = \frac{1}{8}\int_0^4 x^2 dx = \frac{64}{24} = \frac{8}{3} = 2.67.$$

(b) $E(x^2)$

$$= \frac{1}{8}\int_0^4 x^2 \cdot x\,dx = \frac{1}{8}\int_0^4 x^3 dx = 8.$$

(c) $Var(x)$

$$= E(x^2) - (E(x))^2 = 8 - \left(\frac{8}{3}\right)^2 = 8 - \frac{64}{9} = \frac{8}{9} = 0.89$$

(d) $S.D(x)$

$$= \sqrt{Var(x)} = \sqrt{\frac{8}{9}} = 0.942.$$

(e) $Var(3x+2)$

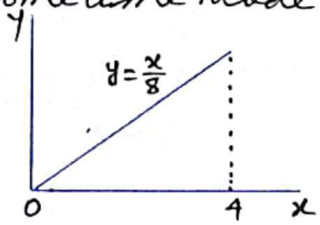
$$= 9.Var(x) = 9 \times \frac{8}{9} = 8$$

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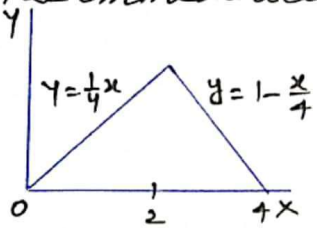
THE MODE

Mode is the value of x for which $f(x)$ is greatest in the given interval of x.

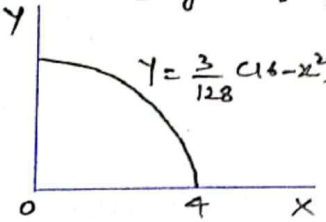
① Sometime Mode can be immediately deduced by the graph.



Mode = 4



Mode = 2



Mode = 0

② Some times for Mode we use second derivative test. Stationary point (i.e $x=a$) will be the Mode.

EXAMPLE ③ If 'x' is a continuous RV having p.d.f

$$f(x)=\begin{cases}\frac{1}{80}(2+x)(4-x), & 0\le x\le 4\\ 0, & \text{otherwise.}\end{cases}$$

Find Mode.

SOLUTION we apply second desivative test for Mode.

Here $f(x)=\frac{1}{80}(8-2x+4x-x^2)$

$=\frac{1}{80}(-x^2+2x+8)$

F.O.C. $f'(x)=0$

$\frac{1}{80}(-2x+2)=0$

$-2x+2=0$

$2x=2 \Rightarrow \boxed{x=1}$

S.O.C $f''(x)=-\frac{2}{80}=-\frac{1}{40}=-ve$

$\Rightarrow f(x)$ has Maximum value at $x=1$

Hence $\boxed{\text{Mode}=1}$

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EXAMPLE-④ A random variable x has p.d.f

$$f(x) = \begin{cases} Ax(6-x)^2, & 0 \le x \le 6 \\ 0, & \text{elsewhere.} \end{cases}$$

Find the following

(a) value of A

(b) Mean

(c) Mode (d) Variance (e) standard deviation

SOLUTION

(a) For the value of A. $\int_0^6 Ax(6-x)^2 dx = 1$

$$108.A = 1$$

$$\boxed{A = \frac{1}{108}}$$

(b) Mean

$$\text{Mean} = E(x) = \int_0^6 x.x(6-x)^2 dx$$

$$= \int_0^6 x^2(6-x)^2 dx = 2.4$$

(c) Mode. Here $f(x) = \frac{1}{108}x(36+x^2-12x) = \frac{1}{108}(36x+x^3-12x^2)$

Now F.O.C $f'(x) = 0 \Rightarrow \frac{1}{108}(36+3x^2-24x) = 0$

$3x^2-24x+36 = 0$, $x^2-8x+12 = 0 \Rightarrow x = 2, x = 6$

Now S.O.C $f''(x) = \frac{1}{108}(6x-24)$

at $x = 2$, $f''(x) = \frac{1}{108}(-12) = -ve \Rightarrow f(x)$ has Max value.

at $x = 6$, $f''(x) = \frac{1}{108}(12) = +ve \Rightarrow f(x)$ has Min value.

So $\boxed{\text{Mode} = 2}$

(d) Variance. $\text{Var}(x) = E(x^2) - (E(x))^2$, Now

$E(x^2) = \frac{1}{108}\int_0^6 x^2.x(6-x)^2 dx = \frac{1}{108}\int_0^6 x^3.(6-x)^2 dx = 7.2$

So $\text{Var}(x) = 7.2 - (2.4)^2 = 1.44$

(e) $S.D(x) = \sqrt{\text{Var}(x)} = \sqrt{1.44} = 1.2$

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EXAMPLE-⑤ The time taken to perform a particular task, t hours, has the probability density function.

$$f(t) = \begin{cases} 10ct^2 & , 0 \le t \le 0.6 \\ 9c(1-t) & , 0.6 \le t \le 1.0 \\ 0 & , \text{otherwise.} \end{cases}$$

where 'c' is a constant.

(a) Find the value of c.

(b) write down the most likely time.

(c) Find the expected time.

(d) Find the probability that time will be more than 48 mins.

(e) " " " " " " b/w 24 and 48 mins.

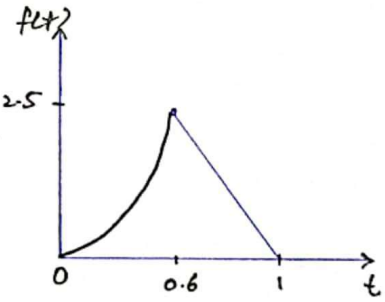
SOLUTION (a) For the value of 'c': $\int_0^1 f(t)\,dt = 1$

$$\text{or } \int_0^{0.6} 10ct^2\,dt + \int_{0.6}^{1} 9c(1-t)\,dt = 1 \Rightarrow c = \frac{25}{36}$$

(b) Mode = ? For Mode Sketch the fn.

From the graph Max value of $t = 0.6$ hrs

$\Rightarrow t = 0.6 \times 60 = 36$ Minutes.



(c) Expected time is

$$E(t) = \int_0^1 t \cdot f(t)\,dt$$

$$= \int_0^{0.6} \frac{125}{18} t \cdot t^2\,dt + \int_{0.6}^{1} \frac{25}{4} \cdot t \cdot (1-t)\,dt = 0.591 \text{ hrs} = 35.5 \text{ mins}$$

(d) $P(t > 48 \text{ mins}) = P(t > 0.8 \text{ hrs}) = \int_{0.8}^{1} 9c(1-t)\,dt$

$$= \frac{25}{4} \int_{0.8}^{1} (1-t)\,dt = 0.125$$

(e) $P(24 < t < 48) = P(0.4 < t < 0.8)$ (Mins → HRS)

$$= \int_{0.4}^{0.6} 10ct^2\,dt + \int_{0.6}^{0.8} 9c(1-t)\,dt = \frac{125}{18}\int_{0.4}^{0.6} t^2\,dt + \frac{25}{4}\int_{0.6}^{0.8} (1-t)\,dt$$

$$= 0.727$$

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